INCH-POUND

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24 January 1994
SUPERSEDING
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27 March 1991

MILITARY SPECIFICATION

PRIMER, ELECTRIC, MARK 45 MOD 1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the minimum requirements for the Electric Primer, Mark 45 Mod 1, herein referred to as the "primer."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the specification (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (Code 8420), Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1390

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-453	Inspection, Radiographic
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications and military standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE CODE 10001 AND 53711)

2434755	Primer, Electric, Mark 45 Mod 1
3208375	Charge, Propelling, 5-Inch, 54 Caliber, Full, Mark 67 Mod 3
5166053	Pack, Primer, Mk 43 Mod 0 Primers, for Gun Cartridges, 5-Inch and Larger
5166057	Shipping Condition, Primer for Gun Cartridges, 5-Inch and Larger

(Application for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordnance Station, Attn: Code 802, Louisville, KY 40214.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE CODE 10001)

OD 49540	Functioning Test of Electric Primers

(Application for copies of OD's should be addressed to: Commanding Officer, Naval Ship Weapons Systems Engineering Station, NAVSEA Data Support Activity (Code 5H30), Port Hueneme, CA 93043.)

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WR-53/884 Palletizing Domestic Unit Load

(Application for copies of WR's should be addressed to: Commanding Officer, Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Characteristics. The primer shall conform to all the requirements specified herein and on Drawing 2434755 and associated documents.

3.3 Radiographic (M101). The primer shall not exhibit any discernible gaps as identified in figure 1 (see 4.7.1).

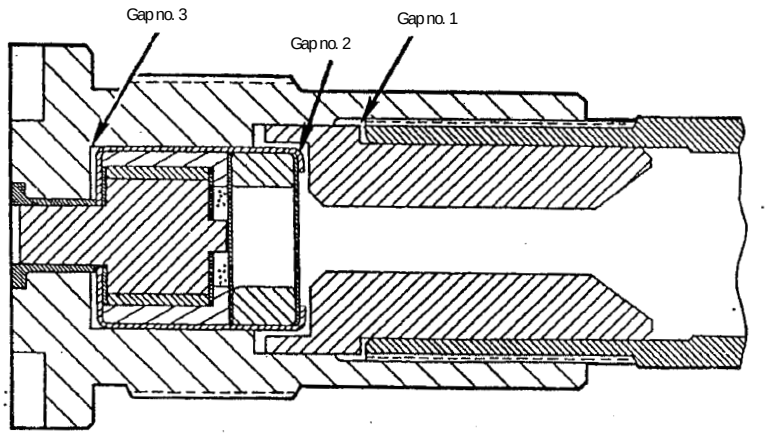


FIGURE 1. Gap identification.

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3.4 Black powder moisture content (M102). The moisture content of the black powder within the primer shall not exceed 0.70 percent by weight (see 4.7.2).

3.5 Detail strip. The primer and all its components shall conform to the following requirements (see 4.7.3):

a. (C1) Each critical characteristic shall be as specified in the applicable documents listed on Drawing 2434755 and this specification.

b. (M103) Each major characteristic shall be as specified in the applicable documents listed on Drawing 2434755 and this specification.

3.6 Electrical resistance (M104). The primer shall have an electrical resistance of 0.10 to 0.18 ohm (see 4.7.4).

3.7 Functional. The primer shall meet the following criteria (see 4.7.5):

a. (M105) Primer functioning time shall be a maximum of 0.006 second;

b. (M106) The primer shall not misfire (see 6.7.1);

c. (M107) The primer shall not exhibit blowback (see 6.7.2); and

d. (C2) The primer shall not exhibit a split stock, split tube, or detachment of metallic components other than blowout of ignition element closing discs.

3.8 Vibration. The primer shall meet the following requirements (see 4.7.6):

a. (C3) The primer shall not function during vibration;

b. (C4) The stock and tube shall maintain material integrity and the primer shall maintain assembly integrity at the mating threads of the stock and tube during vibration;

c. (M108) The primer shall not otherwise physically breakdown during vibration;

d. The primer shall meet the requirements of 3.6 after vibration; and

e. The primer shall meet the requirements of 3.7 after vibration.

3.9 Ballistics. The primer when assembled into a 5"/54 propelling charge shall provide ignition which will meet the criteria of 3.7 b, c, and d, and the following (see 4.7.7):

a. (M109) The ejection time of the projectile from the gun barrel shall not be greater than 0.050 second after application of firing energy.

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3.10 Workmanship. All parts shall be free of chips, dirt, grease, rust, and all other foreign materials. There shall be no burrs or other imperfections of manufacture which would affect the assembly, functioning, or handling of the product. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All markings shall be legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Test and examination methods (see 6.2). Primers shall be selected at random for each test or examination. Identification of each primer's complete test and examination results shall be secured for additional examination until otherwise directed by the contracting activity (see 6.6).

4.4 First article inspection. The first article shall be a homogeneous sample of 130 assembled primers and five each of every component part and subassembly (less explosive components) fabricated, packaged, and packed in accordance with the requirements of this specification and the contract or purchase order using the methods and procedures to be used for production. The first article shall be submitted to the facility designated in the contract or order for test and examination in accordance with the first article inspection plan of table I (see 6.2 and 6.4). Prior to delivery, a first article shall be inspected and tested by the contractor for all requirements specified herein.

4.4.1 First article sample rejection. Failure of the first article sample to meet any of the requirements shall be cause for rejection of the sample.

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TABLE I. First article and quality conformance inspection plan.

Test or Examination	Requirement Paragraph	Test Paragraph	First Article Inspection			Quality Conformance Inspection		
			Sample Size	Accept	Reject	Sample Size	Accept	Reject
Radiographic	3.3	4.7.1.	130[1]	0	1	105[1]	0	1
Electrical resistance	3.6	4.7.4	130[1]	0	1	105[1]	0	1
Functional Functioning time Misfire Blowback Split stock, split tube, or parts detached	3.7 a. b. c d.	4.7.5	30	0 0 0 0	1 1 1 1	25	0 0 0 0	1 1 1 1
Vibration Non-functional Integrity of stock, tube, assembly, etc.	3.8 a b, c. d e.	4.7.6	9	0 0 0 0	1 1 1 1	9	0 0 0 0	1 1 1 1
Ballistic ejection time	3.9	4.7.7	20	0	1			
Detail Strip Critical charac. Major characteristics	3.5	4.7.3	56	0 0	1 1	56	0 1	1 2
Black powder moisture content	3.4	4.7.2	5[2]	0	1	5[2]	0	1
Packaging	Section 5	4.7.8	1 pk	0	1	1 pk	0	1

[1] Radiographic examination and resistance test shall be performed on all samples prior to any other test or examination.
[2] Samples shall be taken from those for the detail strip examination.

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4.5 Quality conformance inspection.

4.5.1 Lot formation. Unless otherwise specified in the contract or purchase order (see 6.2), production lots shall consist of not more than 20,000 homogeneous primers plus those required for inspection (see 4.5.2). Each production lot shall contain only one Class 2 black powder lot and one Class 4 black powder lot and only one tubing heat treat lot.

4.5.2 Inspection. Unless otherwise specified in the contract or purchase order (see 6.2), 105 samples shall be selected at random from each production lot for test and examination in accordance with the quality conformance inspection plan of table I. This sample is in addition to those primers required for contractor inspection and tests.

4.6 Classification of characteristics. The characteristics verified by the tests and examination herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements not annotated with a classification code are classified as minor.

4.7 Methods of inspection.

4.7.1 Radiographic. The primer shall be subjected to a radiographic inspection in accordance with MIL-STD-453. The primer shall be mounted in the vertical position with the primer stock up and X-rayed perpendicular to its axis, focused at a point 0.6 $\pm$ 0.2 inch from the stock face. Any primer which meets the requirements of 3.3 but appears to have been misassembled shall be subjected to detail strip and shall meet the requirements of 3.5.

4.7.2 Determination of moisture content. Moisture analysis procedure for the black powder shall be as follows:

a. Ten grams (10.0 g) of black powder from each sample primer shall be tested in the granular form for moisture content.

b. The black powder shall be removed from primer tubes and placed in a dry container with a tight cover. Black powder shall not be exposed to the atmosphere for more than 3 minutes.

c. Sample containers must be covered and placed in a desiccator until ready to test the moisture content.

d. Carefully weigh 2.0 $\pm$ 0.1 g of sample into a tared weighing dish or container. The sample shall be weighed to the nearest 0.1 mg. Black powder shall not be exposed to the atmosphere for more than 3 minutes.

e. Uncovered samples shall then be placed in an oven at 70 to 75°C. Do not draw vacuum.

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desiccator to cool. After cooling, carefully weigh samples to the nearest 0.1 mg. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{E}{X} \times 100$$

Where:

C = A - B
E = A - D

and

A = Sample weight plus container weight, undried, g
B = Undried container weight, g
C = Undried sample weight, g
D = Sample weight plus container weight, after drying, g
E = Weight difference, after drying, g

4.7.3 Detail strip. The primer shall be disassembled to determine conformance to all the discernible critical and major characteristics specified on Drawing 2434755. All inspections shall be performed at the highest level of assembly possible.

4.7.4 Electrical resistance. The primer electrical resistance shall be measured by using a test circuit which limits the bridge current to 50 milliamperes maximum. The primer stock and the electrode shall be used as terminals. Fifteen to twenty pounds force shall be applied to the electrode during resistance measurement with a probe having a 0.050 $\pm$ 0.005 inch spherical radius. Assure that contact on clean bright surfaces is obtained.

4.7.5 Functional. The primer shall be subjected to the functioning test of OD 49540.

4.7.6 Vibration. The primer shall be subjected to the transportation vibration test of MIL-STD-331, Test B1, section 6.1.2, table B1-4, except that the orientation of each primer during this three-phase test shall be as follows:

a. Phase I: Primer axis horizontal
b. Phase II: Primer axis vertical, primer stock down
c. Phase III: Primer axis vertical, primer stock up

The primer shall be marked with a line in the axial direction at the junction of the stock and tube indexing their initial relative position, in order to determine the integrity after vibration. The primer shall be rigidly mounted to the vibration table at the primer stock. After vibration, the primer shall be subjected to the tests of 4.7.4 and 4.7.5.

4.7.7 Ballistics. The primer shall be assembled into a 5"/54 caliber full service propelling charge assembly in accordance with Drawing 3208375. All components shall be from service acceptable lots. The propellant shall be conditioned at 90 $\pm$ 15•F for 24 hours minimum prior to assembly of the propelling charge. All other components shall be conditioned at 85 $\pm$ 20•F for 24 hours minimum prior to assembly of the propelling charge. The assembled propelling charge shall be conditioned at 90 $\pm$ 5•F for 48 hours minimum immediately prior to firing. Serviceable blind, loaded, and plugged projectiles shall be used. The rounds shall be fired in a 5"/54 caliber electrically fired gun with a first quarter life barrel. Projectile ejection time shall be recorded.

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4.7.8 Inspection of packaging. The packaging, packing, and marking shall be visually examined for conformance to section 5.

5. PACKAGING

5.1 Packaging (Level A). The primer shall be packaged in Primer Pack, Mk 43 Mod 0, Drawing 5166053. Twenty-four units of desiccant, conforming to Type I of MIL-D-3464, shall be placed in each container prior to securing the cover.

5.1.1 Inert components. Components and subassemblies shall be packaged in commercial fiberboard containers meeting the requirements of PPP-B-636. Each container shall incorporate corrugated or solid fiberboard dividers or collars suitable for preventing damage in handling and transportation. The gross weight of the container contents shall not be greater than 70 pounds.

5.2 Packing (Level A). Primers packaged in accordance with 5.1 shall be packed in accordance with Drawing 5166057.

5.3 Palletization. Palletization of primers packed in accordance with 5.2 shall be in accordance with WR-53/884.

5.4 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), all markings shall be in accordance with MIL-STD-l29 and 49 CFR 171-178.

5.5 Ammunition data cards. When specified in the contract or purchase order (see 6.2), one ammunition data card prepared in accordance with MIL-STD-1167 shall be placed in the top of each pack (see 5.1), five shall be forwarded to the contracting activity, and twenty shall go to the ammunition depot receiving the primers. For Navy acquisitions, two data cards shall also go to each of the acitivities listed in 6.6 (see 6.3).

5.6 Ammunition lot number. Ammunition lot number shall be in accordance with MIL-STD-ll68.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory)

6.1 Intended use. The Mk 45 Mod 1 Electric Primer is intended for use as the initiator in the Mk 67 Mods, 5"/54 caliber full propelling charge.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

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c. Whether a first article sample is required (see 3.1).

d. The facility designated to perform first article inspection (see 4.4).

e. The facility designated to perform quality conformance inspection if other than the contractor (see 4.1 and 4.5.2).

f. Inspection lot quantity if other than specified in 4.5.1.

g. Inspection lot sample quantity if other than specified in 4.5.2.

h. Test and examination procedures and equipment to be used (see 4.3, 6.3, and 6.6)

i. Special marking, if required (see 5.4)

j. Ammunition data cards (see 5.5 and 6.3)

k. Safety precautions (see 6.5)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-NDTI-80603	Test Procedure	
5.5	DI-MISC-80043	Ammunition Data Card	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government aprroval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.6 Test procedures and reporting. For Navy acquisitions, the contracting activity should include in each work authorization for first article and quality conformance inspections, the requirement for the preparation and forwarding to the activities listed below of all test and examination procedures prior to performance of any inspection. The work authorization should also include the requirement for the forwarding of all test data in an intelligible form to the activities listed below within five working days of the completion of the tests.

Commander
NAVSURWARCENDIV
(Code 4025)
300 Highway 361
Crane, IN 47522-5001

Commander
NAVSURWARCEN
INDIAN HEAD DIV
(Attn: Code 6210)
101 Strauss Ave.
Indian Head, MD 20640-1035

6.7 Definitions.

6.7.1 Misfire. Misfire is the failure of the main black powder charge of the primer to ignite when the primer is subjected to the appropriate firing source.

6.7.2 Blowback. Blowback is gas leakage around or through the stock end of the primer and is verified by observance during firing or by visual evidence of residue after firing.

6.8 Subject term (key word) listing.

Black powder
Initiator
Propelling charge

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

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Preparing activity:
Navy - OS
Navy - OS
Air Force - 99
(Project No. 1390-N663)

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